

N O R T H E A S T L S O G

A Multi-Source FIA-Proxy Classification of Late-Successional and Old- Growth Forest

Maine, New Hampshire, Vermont, New York | Cardinal HPC | April 2026

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github.com/hobros/lsg-ne

TL;DR — Five Findings

What the regional analysis tells us before we get into the methods

1

Maine has the lowest LSOG share in the Northeast

v5.1 design-based estimate: ME 14.1%, NH 31.2%, VT 28.8%, NY 27.2%

2

Northeast forests are aging — except in Maine

NH +11.1, VT +9.1, NY +8.9 percentage points per decade (1999 to 2023). ME +1.6 pp, borderline significant.

3

Public lands carry 1.5 to 3x the LSOG concentration of private lands

ME federal 36% vs private 14%. NY State+Local has more LS+OG (583 K ac) than ME has across all owners.

4

OG-class plots store 2.6 to 3.6x the live carbon of Not-LSOG plots

ME OG = 139 Mg C / ha. Direct climate-mandate relevance for LD 1529.

5

FIA-proxy and LiDAR LSOG products measure different things

Pingree validation (Phase 9, n=125): aggregate v4 share matches Hagan (16.8 vs 18.8%) but plot-by-plot kappa is random. Skid trails / harvest legacy are visible to LiDAR canopy metrics; tree-level changes are visible to FIA.

PART 01

Methods

Why This Matters

Old-growth forest is rare, valuable, and politically salient — but hard to measure

THE CONTEXT

- **LD 1529 (Maine, 2025)**

Directs DACF to produce a comprehensive LSOG conservation report by Nov 2026, incorporating ecological, carbon, and economic considerations.

- **Hagan et al. 2024 LiDAR mapping**

Identified ~165,000 ha of unmapped old forest in Maine's unorganized territories.

- **Thompson et al. 2026**

Pathways for protecting Maine's remaining old-growth on private commercial timberlands.

- **Pelz et al. 2023 (FEM 549)**

Official USFS old-growth methodology applied to FIA on National Forest System lands.

PROJECT GOALS

1. **Build a defensible FIA proxy**

calibrated against Hagan, ORNL, and Pelz benchmarks.

2. **Extend to NH, VT, NY**

for regional comparison across the Northeast.

3. **Quantify carbon and ownership**

for direct LD 1529 policy use.

4. **Produce wall-to-wall maps**

via TreeMap imputation of FIA plots.

v5.1 Classifier: Six Dimensions, /12 Score

Operational at github.com/holoros/lhog-ne (R/fia_lhog_analysis_v5.r)

1 Large-tree BA

BA in trees DBH ≥ 20 in

1pt: ≥ 40 ft²/ac
2pt: ≥ 80 ft²/ac

2 Stand maturity

STDAGE (or max DBH fallback)

1pt: ≥ 80 yr (or DBH ≥ 24 in)
2pt: ≥ 120 yr

3 Structure

TPA-weighted SD of DBH

1pt: ≥ 5 in
2pt: ≥ 8 in

4 Total stocking

Total live BA

1pt: ≥ 100 ft²/ac
2pt: ≥ 150 ft²/ac

5 Deadwood

Snag TPA, data-driven

1pt: ≥ 75 th percentile
2pt: ≥ 90 th

6 Canopy height

Potapov 2021 GEDI RH95

1pt: ≥ 18 m
2pt: ≥ 25 m

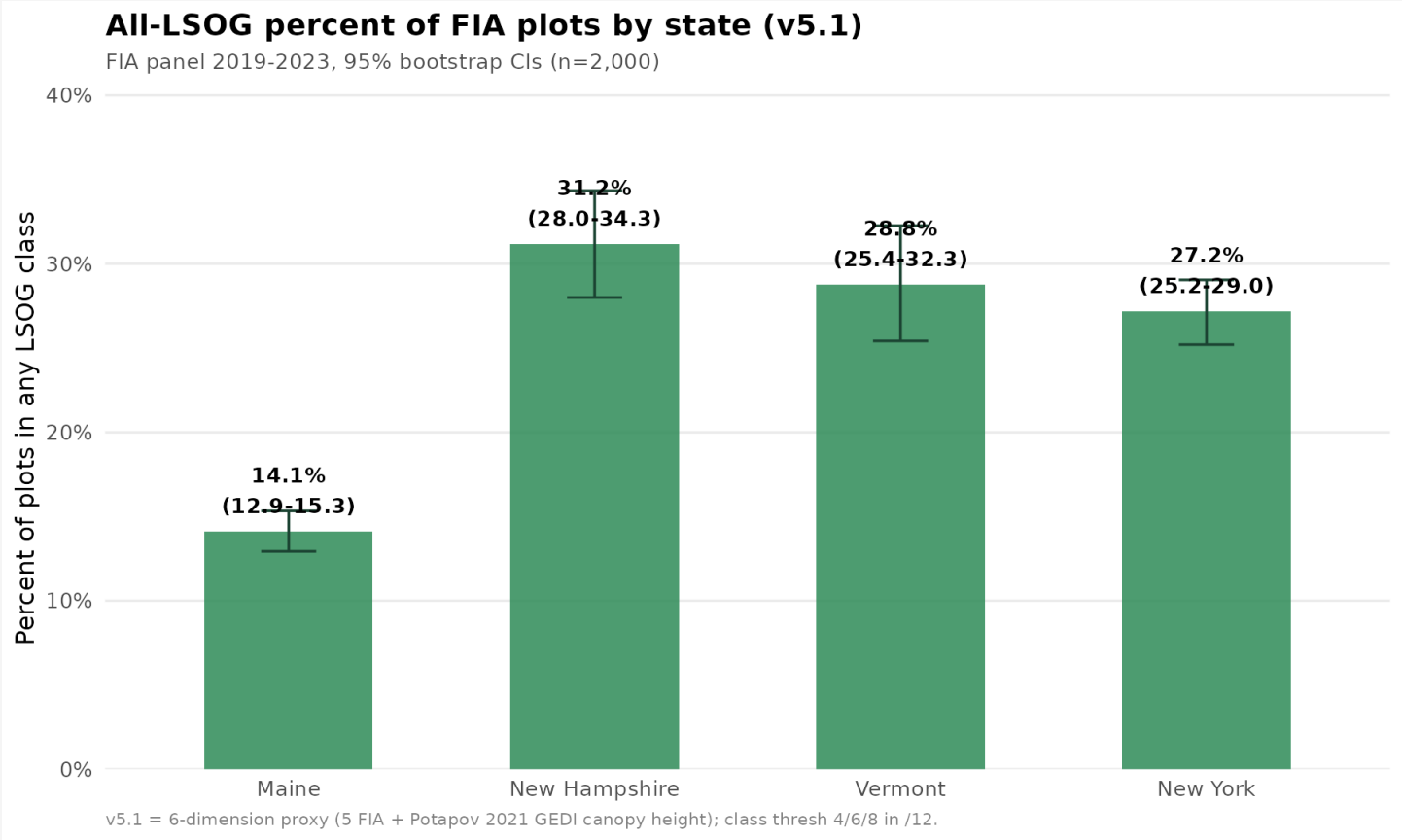
CLASS THRESHOLDS: Transitioning LS ≥ 4 | LS ≥ 6 | OG ≥ 8 (out of 12)

PART 02

Results

Headline Regional Results (FIA panel 2019-2023)

v5.1 plot-based bootstrap estimates with 95% confidence intervals



STATE-LEVEL SHARES

Maine

14.1%

any-LSOG (95% CI 12.9 - 15.3)

New Hampshire

31.2%

any-LSOG (95% CI 28.0 - 34.3)

Vermont

28.8%

any-LSOG (95% CI 25.4 - 32.3)

New York

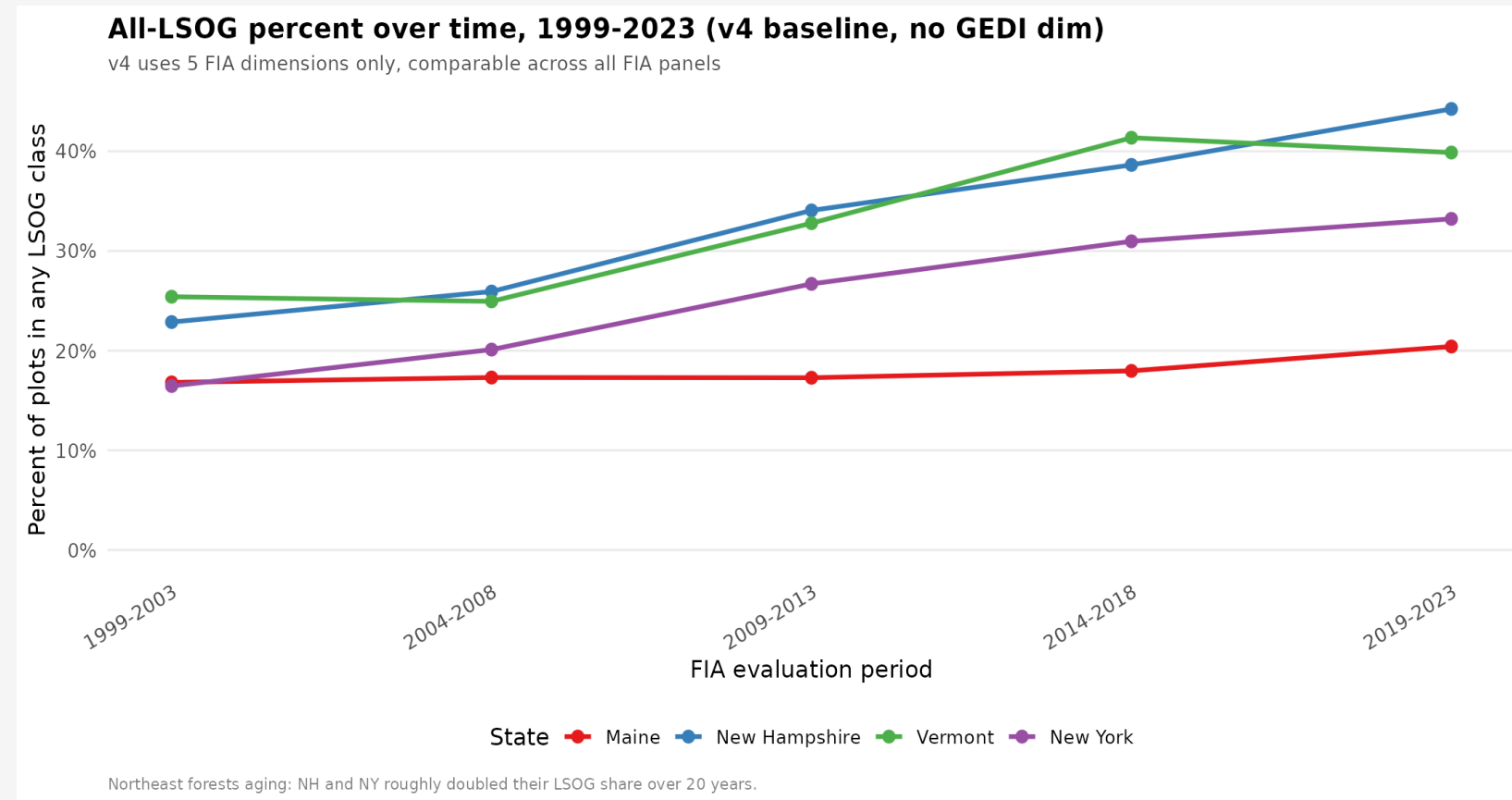
27.2%

any-LSOG (95% CI 25.2 - 29.0)

Maine = lowest LSOG share in NE
despite highest absolute forest area

Forests Are Aging: 20-year LSOG Trend (1999-2023)

v4 baseline (no GEDI), comparable across all 5 FIA panels



TREND TEST

All-LSOG % per decade

NH +11.1 pp

$p < 0.001$

VT +9.1 pp

$p = 0.024$

NY +8.9 pp

$p = 0.002$

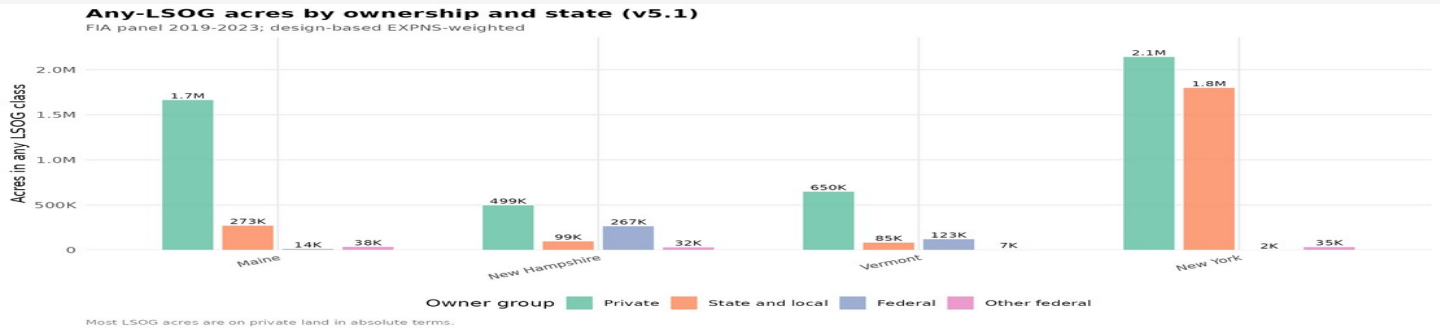
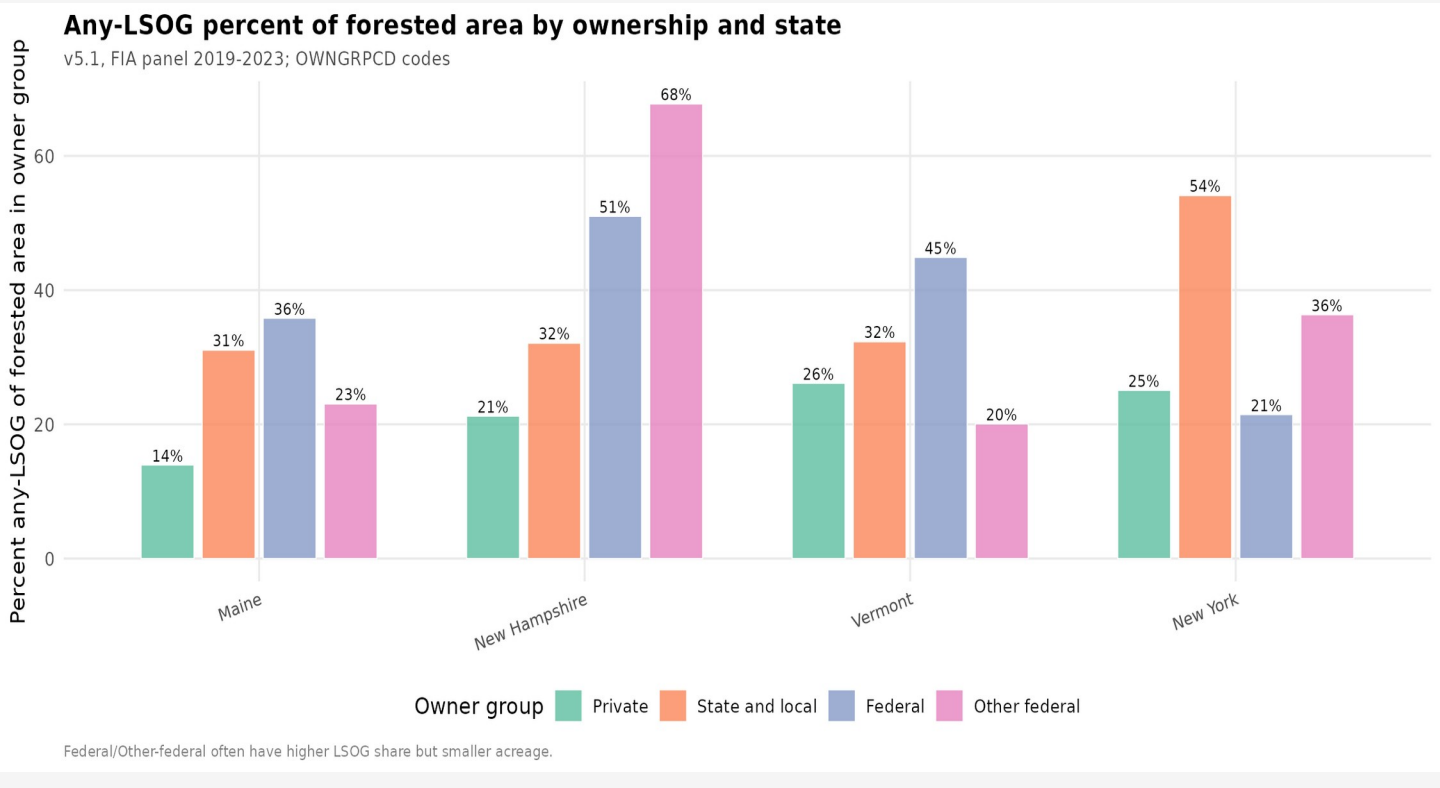
ME +1.6 pp

$p = 0.057$

Maine slope = borderline
(intensive harvest dampens aging signal)

Ownership: Public Lands Have 1.5-3× Higher LSOG Concentration

Design-based EXPNS-weighted estimation, latest panel



KEY POLICY FINDINGS

NY State+Local lands

→ 17.5% LS+OG = 583 K ac
More than ME has across all owners.

Public concentration

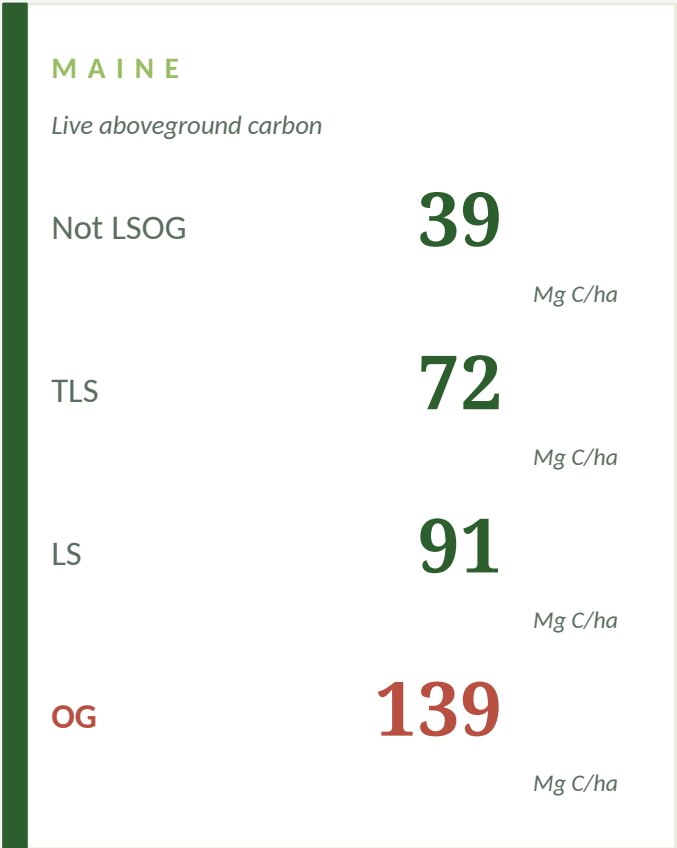
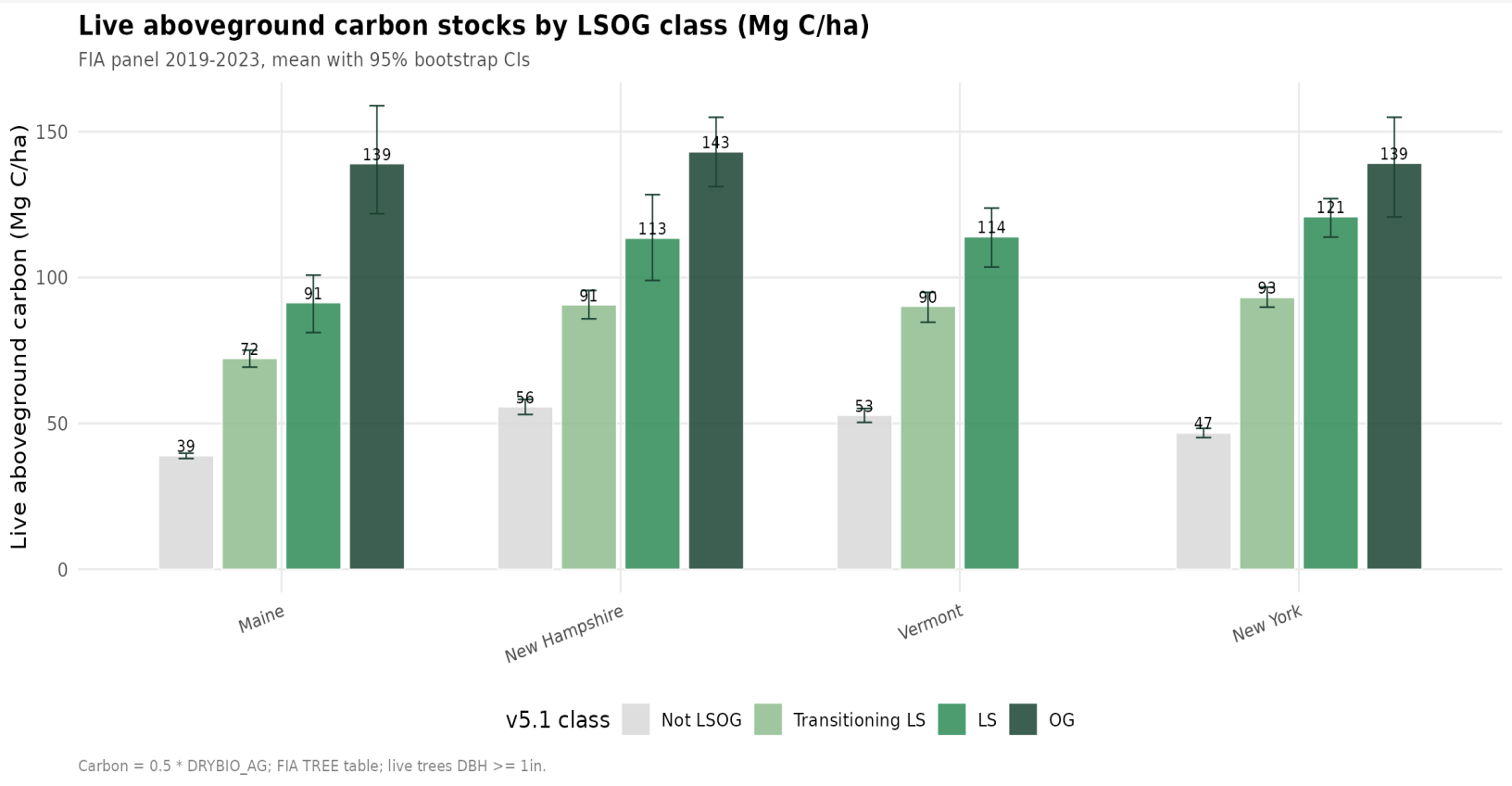
→ ME federal 36% vs private 14%
→ NH federal 51% vs private 21%
→ VT federal 45% vs private 26%

Where total acres live

Private LS+OG total ~987 K ac across NE
the natural target for PES / easements

Carbon: OG Plots Store 2.6-3.6× the Live Carbon of Not-LSOG

Direct LD 1529 climate-mandate relevance



Protecting Maine's ~24 K ac of OG class preserves ~100 Mg C/ha MORE carbon than letting it revert to Not-LSOG. Across the broader 1.99 M ac LSOG pool, that's substantial avoided emissions in policy-relevant terms.

PART 03

Validation and Outlook

Validation Against Independent Products

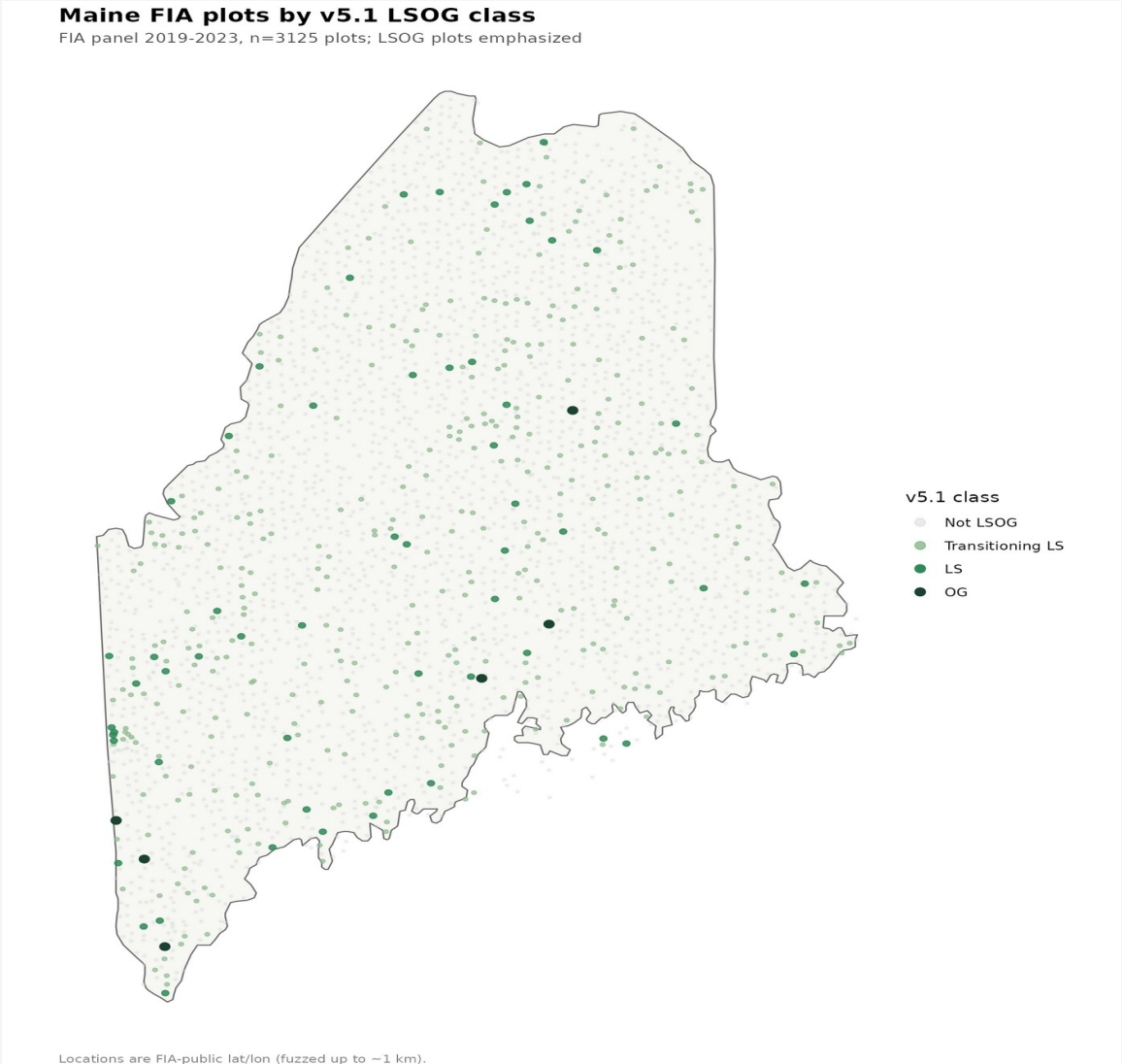
v5.1 fits within range of LiDAR, Bayesian, and USFS-official OG estimates

Source	Coverage	any-LSOG / OG share	Phase 7 kappa vs v5.1
v5.1 (this work)	All ME private + public	ME 14% any LSOG, 0.19% OG	—
Hagan et al. 2024 LiDAR	Maine UT only (9.5 M ac)	21.4% LSOG, 1.0% OG (95 K ac)	BLOCKED (raster pending)
Bruening 2026 (ORNL 2498)	Statewide MOG-prob > 50	~33% mature, 0.13% OG	kappa ~0 on OG-class
Pelz 2023 (FEM 549)	NE NFS lands (~1 M ac)	10.1% Pelz-OG of NFS plots	0.25 LS+OG vs Pelz-OG
TreeMap (this work, v2)	Maine wall-to-wall 30 m	8.0-8.2% any-LSOG	—

Take-away: v5.1 LS+OG combined has fair agreement ($\kappa \approx 0.25$) with Pelz's USFS-official OG criteria. The OG-specific class is poorly validated across all independent products ($\kappa \sim 0$). For policy reporting, LS+OG combined is the most stable category.

Wall-to-Wall: Maine 2020 / 2022 via TreeMap × FIA Imputation

Phase 8 v2 closes the Unknown coverage gap to 0.33% via 1999-2023 panel lookup



MAINE TREEMAP RESULTS

Year	Class	Acres	% pixels
2020	Trans LS	1.25 M	7.5%
2020	LS	118 K	0.7%
2020	Not LSOG	15.30 M	91.5%
2022	Trans LS	1.23 M	7.4%
2022	LS	113 K	0.7%
2022	Not LSOG	15.29 M	91.6%

Why ME 8% < v5.1 plot-based 14%

Phase 8 v2 uses v4-style scoring (no Potapov GEDI canopy height), since the Potapov product is single-year 2019 and not applicable to plots from older FIA panels. Phase 8 v3 (Potapov-aware wall-to-wall) is the natural next step.

Phase 7: Pelz 2023 Cross-Validation on NE NFS Lands

Independent test using USFS-official Eastern Region OG criteria

CONFUSION MATRIX (925 NFS plots in unified table)

v5.1 class	Not Pelz-OG	Pelz-OG	Total
Not LSOG	467	36	503
Transitioning LS	314	49	363
LS	31	24	55
OG	1	3	4
Total	813	112	925

COHEN'S KAPPA

v5.1 OG vs Pelz OG0.044

essentially random

v5.1 LS+OG vs Pelz OG0.253

fair agreement

v5.1 any-LSOG vs Pelz OG0.115

slight

OG-share estimates differ by 25× across products on NE NFS plots

v5.1

0.4%

Hagan UT

1.0%

Pelz R9

10.1%

Phase 9: Seven Islands LiDAR Validation

Pingree Ownership cross-validation against Hagan M2V2b GFW23-masked raster (n = 125 plots)

SHARE-LEVEL COMPARISON (in-extent latest panel, n = 125)

Source	any-LSOG	LS+OG	OG-only
v5.1 (FIA + GEDI)	8.0%	0.8%	0.0%
v4 (FIA, no GEDI)	16.8%	1.6%	0.0%
Hagan at FIA plots	20.0%	1.6%	0.8%
Hagan landscape Pingree	18.8%	2.4%	0.6%

CONFUSION MATRIX v5.1 (rows) x Hagan (cols)

v5.1	Not LS	Trans LS	LS	OG-like
Not LSOG	93	21	1	0
Trans LS	7	1	0	1
LS	0	1	0	0
OG	0	0	0	0

COHEN KAPPA

v5.1 any-LSOG vs Hagan any-LS 0.07

v5.1 LS+OG vs Hagan LS+OG-like -0.01

v4 any-LSOG vs Hagan any-LS -0.01

Aggregate share matches.

Hagan at FIA plots = 20.0%
Hagan landscape Pingree = 18.8%

Plot-by-plot agreement = random

kappa near zero across all class boundaries

FIA plot sample is representative of Pingree (Hagan-at-plots 20.0% closely matches Hagan-landscape 18.8%)

Why FIA-Proxy and LiDAR Disagree at the Plot Level

Skid trails, selective harvest legacy, and what each method can and cannot see

HAGAN LIDAR PROTOCOL (M2V2b)

What it sees:

Eight 1-m LiDAR canopy metrics per hectare:

mean / max / 95th-pct height, rugosity, rumple, cover fractions over 2/6/15 m.

Sensitive to:

Skid trails, harvest gaps, stump removal, light selective harvest.

Canopy disturbance depresses rugosity and cover fractions.

What it misses:

Tree-level changes (large-tree removal, BA reduction, snag dynamics, age) when the residual canopy stays tall.

v5.1 FIA-PROXY CLASSIFIER

What it sees:

FIA tree-level subplot data plus Potapov GEDI 30 m canopy height.

Large-tree BA, total BA, snag TPA, structural SD of DBH, stand age, RH95.

Sensitive to:

Tree-level harvest signal: big trees removed, BA depressed, age clock reset.

Captures change even when canopy structure persists.

What it misses:

Skid-trail and harvest-gap effects within the hectare but outside the FIA subplot. The subplot may not overlap the disturbed pixels.

Both methods are imperfect on heavily managed industrial timberland. They measure genuinely different signals.

Limitations & Caveats

Read every estimate alongside its weakness

1

OG-class precision is low

v5.1 OG kappa ≈ 0 vs Pelz, ORNL, and Hagan. The four v5.1 OG plots may not match the OG-class calls of any independent product. Don't over-interpret state-level OG percentages.

2

FIA-proxy and LiDAR diverge plot-by-plot (Phase 9)

Pingree cross-validation: aggregate share matches (v4 16.8% $\approx$ Hagan 18.8%) but plot-by-plot kappa is random. Skid trails depress Hagan canopy metrics; FIA tree-level data may not record them. Both methods are imperfect on managed timberland.

3

FIA plot fuzzing limits cell-level analysis

Public FIA lat/lon are fuzzed up to ~ 1 km. For 30 m / 100 m raster extractions this introduces small but systematic noise. DUA true coordinates would tighten this; buffered modal sampling does not change kappa estimates.

4

Full Maine UT Hagan raster still not available

Seven Islands shared the Pingree raster (290 K ha, $n=125$ plots). The 4.2 M ha UT raster would expand the LS-class and OG-class samples beyond $n=1$ and $n=1$ respectively in the Hagan reference.

5

ORNL 2498 trains on FIA labels

Bruening et al. 2026 use FIA MOG labels as response variable. Cross-validation is therefore a coherence check, not blind validation.

Project Status and Next Steps

37+ commits on github.com/holoros/lzog-ne; Pingree Hagan validation delivered (Phase 9)

DELIVERED

- v5.1 operational classifier
- Multi-state regional run (ME/NH/VT/NY)
- Bootstrap and FIA design-based CIs
- Ownership, carbon, forest-type breakdowns
- Pelz cross-validation (Phase 7)
- Seven Islands Hagan validation (Phase 9)
- TreeMap NE wall-to-wall + Potapov (Phase 8)
- Manuscript V1 first complete draft

QUEUED UNBLOCKED

- State-specific RH95 thresholds
- Maine UT polygon-based filter
- rFIA design-package cross-validation
- GEDI L2A direct integration
- DUA true-coordinate refinement
- Manuscript V2: reconcile intro numbers
- Extend to MA / CT / RI

BLOCKED / PARTIAL

- Full Maine UT Hagan raster
Pingree available via Seven Islands
4.2 M ha UT-wide still pending
- Acquisition routes:
>> *Our Climate Common*
>> *LD 1529 / DACF working group*